AI ASSISTED CODING

Assignment-15.1

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Task 1 – Student Records API

Task:

Use AI to build a RESTful API for managing student records.

Instructions:

• Endpoints required:

o GET /students → List all students

o POST /students → Add a new student

o PUT /students/{id} → Update student details

o DELETE /students/{id} → Delete a student record

• Use an in-memory data structure (list or dictionary) to

store records.

• Ensure API responses are in JSON format.

Expected Output:

• Working API with CRUD functionality for student records.

Commands + Code + Text Run all

[ ]

Start coding or generate with AI.

[1] 5s

```
pip install fastapi uvicorn
```

```
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvic
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicor
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyic
```

I've installed `fastapi` and `uvicorn`. Now, let's define the API. I'll use `pydantic` for data validation and `fastapi` for the API endpoints. The student records will be stored in a simple Python dictionary in memory.

[2] 0s

```python
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List

app = FastAPI()

# In-memory data store for students
students_db: Dict[int, BaseModel] = {}
next_student_id = 1

# Pydantic model for a student
class Student(BaseModel):
    name: str
    age: int
    major: str

# Pydantic model for updating a student (all fields optional)
class StudentUpdate(BaseModel):
    name: str = None
    age: int = None
    major: str = None

@app.get("/students", response_model=List[Student])
def list_students():
```

[2]
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```python
def list_students():
    """List all students."""
    return list(students_db.values())

@app.post("/students", response_model=Student, status_code=201)
def add_student(student: Student):
    """Add a new student."""
    global next_student_id
    students_db[next_student_id] = student
    student_with_id = student.copy(update={"id": next_student_id})
    next_student_id += 1
    return student_with_id

@app.put("/students/{student_id}", response_model=Student)
def update_student(student_id: int, student_update: StudentUpdate):
    """Update student details."""
    if student_id not in students_db:
        raise HTTPException(status_code=404, detail="Student not found")

    existing_student = students_db[student_id]
    update_data = student_update.dict(exclude_unset=True)
    updated_student = existing_student.copy(update=update_data)
    students_db[student_id] = updated_student
    return updated_student
```

```python
    return updated_student

@app.delete("/students/{student_id}", status_code=204)
def delete_student(student_id: int):
    """Delete a student record."""
    if student_id not in students_db:
        raise HTTPException(status_code=404, detail="Student not found")
    del students_db[student_id]
    return {"message": "Student deleted successfully"}
```

Now that the API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. Also, the API will be accessible via `localhost:8000`. You can test it by running this cell and then using `curl` or a tool like Postman/Insomnia in a separate terminal or web browser.

To interact with the API, you can use the following in a new cell or a separate terminal:

- **GET all students:**

```
!curl http://localhost:8000/students
```

- **POST a new student:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"name": "Alice", "age": 20, "major":
```

- **PUT (update) a student (assuming ID 1 exists):**

```
!curl -X PUT -H "Content-Type: application/json" -d '{"age": 21}' http://localhost:8000/stud
```

- **DELETE a student (assuming ID 1 exists):**

```
!curl -X DELETE http://localhost:8000/students/1
```

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Commands + Code + Text Run all

[4] 0s

```python
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
```

[4] 0s

```python
    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Run the server in a separate task using asyncio.create_task
# This allows the cell to continue executing after starting the server
# and prevents the RuntimeError previously encountered.

# To start the server, call:
# await stopper.start()
# To stop the server, call:
# stopper.stop()

# We will start it now for the user.
await stopper.start()
```

```
INFO:     Started server process [1995]
INFO:     Waiting for application startup.
API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000
```

Task 2 – Library Book Management API

Task:

Develop a RESTful API to handle library books.

Instructions:

• Endpoints required:

o GET /books → Retrieve all books

o POST /books → Add a new book

o GET /books/{id} → Get details of a specific book

o PATCH /books/{id} → Update partial book details

(e.g., availability)

o DELETE /books/{id} → Remove a book

• Implement error handling for invalid requests.

Expected Output:

• Functional API with CRUD + partial updates

Commands + Code + Text Run all

[1]
4s

```
pip install fastapi uvicorn
```

```
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packa
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Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvic
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicor
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyic
```

I've installed `fastapi` and `uvicorn`. Now, let's define the API for managing library books. I'll use `pydantic` for data validation and `fastapi` for the API endpoints. The book records will be stored in a simple Python dictionary in memory.

[2]
0s

```python
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional

app = FastAPI(title="Library Book Management API")

# In-memory data store for books
books_db: Dict[int, BaseModel] = {}
next_book_id = 1

# Pydantic model for a book
class Book(BaseModel):
    title: str
    author: str
    publication_year: int
    is_available: bool = True
    id: Optional[int] = None # Optional ID for response, will be assigned by API

# Pydantic model for updating a book (all fields optional)
class BookUpdate(BaseModel):
    title: Optional[str] = None
    author: Optional[str] = None
    publication_year: Optional[int] = None
    is_available: Optional[bool] = None
```

[2]
0s

```python
@app.get("/books", response_model=List[Book], summary="Retrieve all books")
def get_all_books():
    """Retrieve a list of all books in the library."""
    return list(books_db.values())

@app.post("/books", response_model=Book, status_code=201, summary="Add a new book")
def add_book(book: Book):
    """Add a new book to the library.

    - **title**: The title of the book.
    - **author**: The author of the book.
    - **publication_year**: The year the book was published.
    - **is_available**: Whether the book is currently available (defaults to True).
    """
    global next_book_id
    book.id = next_book_id
    books_db[next_book_id] = book
    next_book_id += 1
    return book

@app.get("/books/{book_id}", response_model=Book, summary="Get details of a specific book")
def get_book_details(book_id: int):
    """Get details of a specific book by its ID."""
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")
```

[2]
0s

```python
@app.patch("/books/{book_id}", response_model=Book, summary="Update partial book details")
def update_book_details(book_id: int, book_update: BookUpdate):
    """Update partial details of a book by its ID.

    - **title**: New title (optional).
    - **author**: New author (optional).
    - **publication_year**: New publication year (optional).
    - **is_available**: New availability status (optional).
    """
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")

    existing_book = books_db[book_id]
    update_data = book_update.dict(exclude_unset=True) # Only update provided fields
    updated_book_data = existing_book.dict()
    updated_book_data.update(update_data)
    updated_book = Book(**updated_book_data)
    books_db[book_id] = updated_book
    return updated_book

@app.delete("/books/{book_id}", status_code=204, summary="Remove a book")
def delete_book(book_id: int):
    """Remove a book from the library by its ID."""
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")
```

```
[2]
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        del books_db[book_id]
        return {"message": "Book deleted successfully"}
```

Now that the API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. Also, the API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal:

- **GET all books:**

```
!curl http://localhost:8000/books
```

- **POST a new book:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"title": "The Hobbit", "author": "J.R
```

- **GET a specific book (e.g., ID 1):**


```
!curl -X POST -H "Content-Type: application/json" -d '{"title": "The Hobbit", "author": "J.R
```

- **GET a specific book (e.g., ID 1):**


```
!curl http://localhost:8000/books/1
```

- **PATCH (update) book details (e.g., change availability for ID 1):**

```
!curl -X PATCH -H "Content-Type: application/json" -d '{"is_available": false}' http://locall
```

- **DELETE a book (e.g., ID 1):**

```python
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Library Book Management API is running at http://0.0.0.0:8000 or http://127.0.0
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")
```

```python
    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Start the server
await stopper.start()
```

```
INFO:     Started server process [3359]
INFO:     Waiting for application startup.
Library Book Management API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000
```

Task 3 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

• Endpoints required:

o GET /employees → List all employees

o POST /employees → Add a new employee with

salary details

o PUT /employees/{id}/salary → Update salary of an

employee

o DELETE /employees/{id} → Remove employee

from system

• Use AI to:

o Suggest data model structure.

o Add comments/docstrings for all endpoints.

Expected Output:

• API supporting salary management with clear

documentation.

[1]
8s

```
pip install fastapi uvicorn
```

```
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvic
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicor
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio
```

I've ensured `fastapi` and `uvicorn` are installed. Now, let's define the data models for the Employee Payroll API.

Commands + Code + Text Run all

Suggested Data Model Structure

I recommend using Pydantic models for data validation and serialization.

- `Employee` **Model:** This will represent an employee, including their name, position, and salary. The `id` will be optional for input but assigned by the API.
- `SalaryUpdate` **Model:** This will specifically handle updates to an employee's salary.

The employee records will be stored in an in-memory dictionary for simplicity.

[2]
0s

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional

app = FastAPI(title="Employee Payroll API")

# In-memory data store for employees
employees_db: Dict[int, BaseModel] = {}
next_employee_id = 1

# Pydantic model for an Employee
class Employee(BaseModel):
    name: str
    position: str
```

```python
class Employee(BaseModel):
    name: str
    position: str
    salary: float
    id: Optional[int] = None  # Optional ID for response, will be assigned by API

# Pydantic model for updating only the salary
class SalaryUpdate(BaseModel):
    salary: float

@app.get("/employees", response_model=List[Employee], summary="List all employees")
def list_employees():
    """Retrieve a list of all employees in the system, including their details."""
    return list(employees_db.values())

@app.post("/employees", response_model=Employee, status_code=201, summary="Add a new employee"
def add_employee(employee: Employee):
    """Add a new employee to the system with their name, position, and salary.

    - **name**: The full name of the employee.
    - **position**: The job title or position of the employee.
    - **salary**: The current salary of the employee.
    """
```

```python
    global next_employee_id
    employee.id = next_employee_id
    employees_db[next_employee_id] = employee
    next_employee_id += 1
    return employee

@app.put("/employees/{employee_id}/salary", response_model=Employee, summary="Update salary of
def update_employee_salary(employee_id: int, salary_update: SalaryUpdate):
    """Update the salary of a specific employee identified by their ID.

    - **employee_id**: The unique identifier of the employee.
    - **salary**: The new salary value for the employee.
    """
    if employee_id not in employees_db:
        raise HTTPException(status_code=404, detail="Employee not found")

    existing_employee = employees_db[employee_id]
    existing_employee.salary = salary_update.salary
    employees_db[employee_id] = existing_employee
    return existing_employee

@app.delete("/employees/{employee_id}", status_code=204, summary="Remove an employee")
def delete_employee(employee_id: int):
    """Remove an employee record from the system by their ID.
```

```
    """
    if employee_id not in employees_db:
        raise HTTPException(status_code=404, detail="Employee not found")
    del employees_db[employee_id]
    return {"message": "Employee deleted successfully"}
```

Now that the Employee Payroll API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. The API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal:

- **GET all employees:**

```
!curl http://localhost:8000/employees
```

- **POST a new employee:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"name": "John Doe", "position": "Softu
```

- **PUT (update) an employee's salary (e.g., for employee ID 1):**

```
!curl -X PUT -H "Content-Type: application/json" -d '{"salary": 80000.0}' http://localhost:8
```

- **DELETE an employee (e.g., employee ID 1):**

```
!curl -X DELETE http://localhost:8000/employees/1
```

```
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Employee Payroll API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
```

```
[3]
0s
        print("Employee Payroll API is running at http://0.0.0.0:8000 or
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Start the server
await stopper.start()

INFO:     Started server process [4376]
INFO:     Waiting for application startup.
Employee Payroll API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000
```

Task 4 – Real-Time Application: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

• Endpoints required:

o GET /menu → List available dishes

o POST /order → Place a new order

o GET /order/{id} → Track order status

o PUT /order/{id} → Update an existing order (e.g.,

change items)

o DELETE /order/{id} → Cancel an order

• AI should generate:

o REST API code

o Suggested improvements (like authentication,

pagination)

Expected Output:

• Fully working API simulating a food ordering backend.

Deliverables (For All Tasks)

1. AI-generated prompts for code and test case generation.

2. At least 3 assert test cases for each task.

3. AI-generated initial code and execution screenshots.

4. Analysis of whether code passes all tests.

5. Improved final version with inline comments and explanation.

6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.

[1] 6s

```
pip install fastapi uvicorn
```

```
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvic
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicor
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyic
```

I've ensured `fastapi` and `uvicorn` are installed. Now, let's define the API for the online food ordering system. I'll use `pydantic` for data validation and `fastapi` for the API endpoints. Menu items and orders will be stored in in-memory dictionaries.

[2] 0s

```python
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional
import uuid

app = FastAPI(title="Online Food Ordering API")

# In-memory data store for menu items and orders
menu_db: Dict[str, BaseModel] = {}
orders_db: Dict[str, BaseModel] = {}

# Initialize some sample menu items
menu_db["dish1"] = {"name": "Pizza Margherita", "description": "Classic pizza with tomato and
menu_db["dish2"] = {"name": "Spaghetti Carbonara", "description": "Pasta with eggs, hard chees
menu_db["dish3"] = {"name": "Caesar Salad", "description": "Fresh romaine lettuce with Caesar

# Pydantic model for a Dish
class Dish(BaseModel):
    name: str
    description: str
    price: float
    dish_id: Optional[str] = None # Added for clarity, will be key in menu_db

# Pydantic model for an item within an Order
class OrderItem(BaseModel):
```

```python
# Pydantic model for an item within an Order
class OrderItem(BaseModel):
    dish_id: str
    quantity: int

# Pydantic model for an Order
class Order(BaseModel):
    items: List[OrderItem]
    status: str = "pending"
    total_price: float = 0.0
    order_id: Optional[str] = None # Will be generated by the API

# Pydantic model for updating an existing order
class OrderUpdate(BaseModel):
    items: Optional[List[OrderItem]] = None
    status: Optional[str] = None


@app.get("/menu", response_model=List[Dish], summary="List available dishes")
def get_menu():
    """Retrieve a list of all available dishes on the menu."""
    # Convert menu_db values to Dish models and add dish_id
    dishes = []
    for dish_id, dish_data in menu_db.items():
        dish_obj = Dish(**dish_data)
```

[2]
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```python
        dishes.append(dish_obj)
    return dishes

@app.post("/order", response_model=Order, status_code=201, summary="Place a new order")
def place_order(order: Order):
    """Place a new order with a list of dishes and quantities.

    - **items**: A list of `OrderItem` objects, each containing `dish_id` and `quantity`.
    """
    order_id = str(uuid.uuid4()) # Generate a unique order ID
    total_price = 0.0

    # Calculate total price and validate dishes
    for item in order.items:
        if item.dish_id not in menu_db:
            raise HTTPException(status_code=404, detail=f"Dish '{item.dish_id}' not found in m
        dish_price = menu_db[item.dish_id]["price"]
        total_price += dish_price * item.quantity

    order.order_id = order_id
    order.total_price = total_price
    orders_db[order_id] = order
    return order

@app.get("/order/{order_id}", response_model=Order, summary="Track order status")
```

```
[2]
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def get_order_status(order_id: str):
    """Track the status and details of a specific order by its ID."""
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")
    return orders_db[order_id]

@app.put("/order/{order_id}", response_model=Order, summary="Update an existing order")
def update_order(order_id: str, order_update: OrderUpdate):
    """Update an existing order's items or status.
    Note: This allows for significant changes, in a real app this might be more restricted.

    - **items**: An optional list of `OrderItem` objects to replace current items.
    - **status**: An optional string to update the order status (e.g., 'preparing', 'delivered
    """
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")

    existing_order = orders_db[order_id]
    update_data = order_update.dict(exclude_unset=True)

    # Update items if provided
    if "items" in update_data:
        new_total_price = 0.0
        for item in update_data["items"]:
            if item["dish_id"] not in menu_db:
```

```
[2]
✓ 0s
            if item["dish_id"] not in menu_db:
                raise HTTPException(status_code=404, detail=f"Dish '{item['dish_id']}' not fou
            dish_price = menu_db[item["dish_id"]]["price"]
            new_total_price += dish_price * item["quantity"]
        existing_order.items = [OrderItem(**item) for item in update_data["items"]]
        existing_order.total_price = new_total_price

    # Update status if provided
    if "status" in update_data:
        existing_order.status = update_data["status"]

    orders_db[order_id] = existing_order
    return existing_order

@app.delete("/order/{order_id}", status_code=204, summary="Cancel an order")
def cancel_order(order_id: str):
    """Cancel an existing order by its ID."""
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")

    # In a real system, you might change status to 'cancelled' instead of deleting
    del orders_db[order_id]
    return {"message": "Order cancelled successfully"}
```

Now that the Online Food Ordering API is defined, you can run it using Uvicorn. The API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal. Note that `order_id`s are UUIDs and will be generated by the `POST /order` endpoint.

- **GET menu:**

```
!curl http://localhost:8000/menu
```

- **POST a new order:**

```
!curl -X POST -H "Content-Type: application/json" -d '[{"dish_id": "dish1", "quantity": 2},
```

(Replace `dish_id` with actual IDs from `/menu`)

- **GET order status (replace `YOUR_ORDER_ID`):**

```
!curl http://localhost:8000/order/YOUR_ORDER_ID
```

[3]
✓ 0s

```
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Online Food Ordering API is running at http://0.0.0.0:8000 or http://127.0.0.1:
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
```

```python
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Online Food Ordering API is running at http://0.0.0.0:8000 or http://127.0.0.
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Start the server
await stopper.start()
```

```
INFO:     Started server process [5333]
INFO:     Waiting for application startup.
Online Food Ordering API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000
```